SE EXPERIMENT 8

Aim: To prepare RMMM plan for the Smart Parking Management System (SPMS).

Theory:

Planning the risk management

The proactive strategy for risk estimation is used which helps us in identifying the possible threats that can occur during the project well in advance. Accordingly, steps to avoid, monitor and manage the risk are to be carried out and noted down in the form of RMMM plan.

Example:

THE RMMM PLAN

A risk management strategy can be included in the software project plan or the risk management steps can be organized into a separate Risk Mitigation, Monitoring and Management Plan. The RMMM plan documents all work performed as part of risk analysis and is used by the project manager as part of the overall project plan. Some software teams do not develop a formal RMMM document. Rather, each risk is documented individually using a risk information sheet (RIS). In most cases, the RIS is maintained using a database system, so that creation and information entry, priority ordering, searches, and other analysis may be accomplished easily.

Once RMMM has been documented and the project has begun, risk mitigation and monitoring steps commence. As we have already discussed, risk mitigation is a problem avoidance activity. Risk monitoring is a project tracking activity with three primary objectives:

1. To assess whether predicted risks do, in fact, occur.
2. To ensure that risk aversion steps defined for the risk are being properly applied.
3. To collect information that can be used for future risk analysis.

In many cases, the problems that occur during a project can be traced to more than one risk. Another job of risk monitoring is to attempt to allocate origin (what risk(s) caused which problems throughout the project).

Identifying Risks:

Risk ID	Risk Summary	Probability	Impact	Risk Exposure
1	Gate Barrier Lockup Trapping Vehicles	40%	1	Rs. 64,000/month
2	Concurrent Slot Double-Booking in Redis	65%	2	Rs. 52,000/month
3	QR Code Spoofing & Revenue Leakage	55%	2	Rs. 49,500/month
4	Schedule Slippage Beyond 3-Month Constraint	75%	3	Rs. 41,250/month

Impact:

1 – Catastrophic
2 – Critical
3 – Marginal
4 – Negligible

Risk Exposure:

Risk: Gate Barrier Lockup Trapping Vehicles
The automated gate barrier system relies on microcontroller units (ESP32/Raspberry Pi) and optical scanners receiving open/close signals via MQTT. In the event of a system hang, power fluctuation, or network drop, the barrier arm could remain locked shut. This traps incoming and departing vehicles inside or outside the facility, completely halting multi-storey parking flow and causing severe traffic backlogs on connecting city streets.

Risk Probability: 80%

Risk Impact:
The risk impact is catastrophic (Scale 1) as physical vehicle entrapment directly violates basic facility safety requirements (SR-1), incurs commercial legal liabilities, and halts revenue operations completely.
The risk can be mitigated by installing spring-loaded fail-safe hardware relays that disengage the barrier lock into manual swing mode upon loss of heartbeat signals, alongside dedicated UPS units and emergency manual override switches manned by attendants. An on-site electrical support and triage setup costs Rs. 1,60,000/month.

Risk Exposure: $\text{RE} = 0.40 \times \text{Rs. } 1{,}60{,}000 = \text{Rs. } 64{,}000/\text{month}$

Risk Information Sheet

Risk ID: 1	**Date:** 12/09/2026	**Probability:** 40%	**Impact:** Catastrophic
Description: Microcontroller barriers or MQTT broker links crash, locking physical barrier arms shut and trapping vehicles within the facility.			
Refinement/Context: **Sub condition 1:** MQTT messaging drops or power failures leave microcontrollers unresponsive. **Sub condition 2:** Entry and exit blockages back up traffic onto access roads, creating critical safety hazards.			
Mitigation/Monitoring: 1. Implement fail-safe relays defaulting barriers to manual mode during network/power outage (SR-1). 2. Deploy uninterruptible power supply (UPS) batteries and mechanical manual overrides at every gate. 3. Monitor heartbeat telemetry every 5 seconds.			
Management: RE computed to be Rs. 64,000 per month.			
Current Status: Mitigation steps to be initiated.			
Originator: Aditya Patil		**Assigned:** Aryan Punalekar	

Risk Exposure:

Risk: Concurrent Slot Double-Booking in Redis
The system provides advance slot reservation with an interactive floor grid, placing a temporary 15-minute hold on a slot while checkout proceeds. When multiple drivers target the same high-demand slot simultaneously during peak arrival rushes, a race condition may cause the system to confirm the same physical parking spot for two vehicles.

Risk Probability: 65%

Risk Impact:
The risk impact is critical (Scale 2) as arriving motorists find their reserved space occupied by another car, degrading trust, creating facility bottlenecks, and generating operational disputes.

The risk can be mitigated by utilizing Redis atomic distributed locking (SETNX) with a strict 15-minute Time-To-Live (TTL) alongside PostgreSQL serializable transaction isolation. Dedicated database concurrency optimization and cloud caching services cost Rs. 80,000/month.

Risk Exposure: RE $= 0.65 \times$ Rs. 80,000 $=$ Rs. 52,000/month

Risk Information Sheet

Risk ID: 2	**Date:** 12/09/2026	**Probability:** 65%	**Impact:** Critical
Description: Concurrent reservation requests cause identical parking slots to be locked and confirmed for multiple motorists simultaneously.			
Refinement/Context: **Sub condition 1:** Non-atomic database writes during checkout allow overlapping allocations. **Sub condition 2:** Expired temporary locks fail to release promptly, distorting live availability grids.			
Mitigation/Monitoring: 1. Prevent double-booking by locking slots atomically in Redis during the checkout window. 2. Enforce atomic isolation on transactional database records. 3. Implement automated rollback triggers on failed checkouts.			
Management: RE computed to be Rs. 52,000 per month.			
Current Status: Mitigation steps to be initiated.			
Originator: Aryan Punalekar		**Assigned:** Yajat Padmavat	

Risk Exposure:

Risk: QR Code Spoofing & Revenue Leakage
Contactless gate access operates via optical scanning of digital QR passes generated within the mobile app. Fraudulent motorists could share static QR screenshots or reuse passes to gain unbilled access or evade overstay penalty fees.

Risk Probability: 55%

Risk Impact:
The risk impact is critical (Scale 2) because unauthorized usage directly degrades lot revenue, produces billing discrepancies, and compromises facility security.

The risk can be mitigated by encoding short-lived cryptographic timestamps (TOTP) into QR passes, ensuring tokens expire within 30 seconds and reject static screenshots, paired with optical gate induction loop sensors to permit only one vehicle cycle per validated scan. Anti-fraud verification infrastructure and gateway security cost Rs. 90,000/month.

Risk Exposure: RE $= 0.55 \times$ Rs. 90,000 $=$ Rs. 49,500/month

Risk Information Sheet

Risk ID: 3	**Date:** 12/09/2026	**Probability:** 55%	**Impact:** Critical
Description: Replay of static QR screenshots or pass duplication results in unauthorized lot entry, unbilled stays, and revenue leakage.			
Refinement/Context: **Sub condition 1:** Static barcode screenshots are shared across multiple vehicle drivers. **Sub condition 2:** Lack of synchronized check-out records allows vehicles to evade escalated hourly penalty rates.			
Mitigation/Monitoring: 1. Issue dynamically refreshing QR codes embedded with cryptographic timestamps (REQ-1). 2. Integrate optical scanners directly with barrier ground induction loops to prevent pass tailgating. 3. Log entry/exit timestamps cryptographically in PostgreSQL.			
Management: RE computed to be Rs. 49,500 per month.			
Current Status: Mitigation steps to be initiated.			
Originator: Yajat Padmavat		**Assigned:** Aditya Patil	

Risk Exposure:

Risk: Schedule Slippage Beyond 3-Month Constraint
Project design and implementation constraints mandate delivery within 3 months (CO-1). Hardware-software integration across microcontrollers, ultrasonic sensors, and full-duplex WebSockets introduces unforeseen protocol bugs that threaten the delivery timeline.

Risk Probability: 75%

Risk Impact:
The risk impact is marginal (Scale 3) relative to total facility failure, but leads to team burnout, budget overruns, and delayed commissioning.

The risk can be mitigated by constructing simulated virtual hardware APIs (MQTT emulators) allowing frontend and backend developers to build core logic in parallel without awaiting physical wiring. Automated CI/CD pipelines enforcing 80% test coverage minimize integration friction. The team sprint augmentation and testing tools cost Rs. 55,000/month.

Risk Exposure: RE $= 0.75 \times$ Rs. 55,000 $=$ Rs. 41,250/month

Risk Information Sheet

Risk ID: 4	**Date:** 12/09/2026	**Probability:** 75%	**Impact:** Marginal
Description: Friction during IoT sensor hardware and cloud backend integration causes delays exceeding the 3-month schedule cap.			
Refinement/Context: **Sub condition 1:** Hardware-level MQTT messaging issues consume developer bandwidth, stalling core app features. **Sub condition 2:** Delays in end-to-end testing reduce unit and integration coverage below the required 80% baseline.			
Mitigation/Monitoring: 1. Build virtual hardware emulators to decouple software engineering from on-site sensor delivery. 2. Execute bi-weekly Agile sprints with strict burndown tracking. 3. Enforce 80% code coverage in CI/CD pipelines before merging modules.			
Management: RE computed to be Rs. 41,250 per month.			
Current Status: Mitigation steps to be initiated.			
Originator: Aditya Patil		**Assigned:** Yajat Padmavat	

Annual Risk Management & Mitigation Budget

Risk ID	Strategy / Budget Category	Monthly Cost (Rs.)	Annual Cost (Rs.)
R1	Fail-safe mechanical relays, emergency UPS, manual override staffing	Rs. 18,000	Rs. 2,16,000
R2	High-availability Redis caching tier & distributed lock management	Rs. 14,000	Rs. 1,68,000
R3	Cryptographic dynamic QR licensing & anti-fraud security monitoring	Rs. 12,000	Rs. 1,44,000
R4	CI/CD automation, testing frameworks, and virtual hardware test harnesses	Rs. 11,000	Rs. 1,32,000
Total	Risk Management & Mitigation Reserve	Rs. 55,000	Rs. 6,60,000

Annual Operational Maintenance Budget

Category	Component Description	Monthly Cost (Rs.)	Annual Cost (Rs.)
Cloud Hosting & Infrastructure	Managed AWS EC2 instances, PostgreSQL database, and S3 receipt storage	Rs. 25,000	Rs. 3,00,000
IoT Sensor & Barrier Upkeep	Ultrasonic sensor calibration, lens cleaning, and gate barrier motor servicing	Rs. 18,000	Rs. 2,16,000
Application & API Maintenance	Bug fixes, Node.js security patches, dependency upgrades, and API maintenance	Rs. 30,000	Rs. 3,60,000
Network & Telemetry Connectivity	Broadband connections and 4G IoT SIM cards for entrance/exit barrier controllers	Rs. 7,000	Rs. 84,000
Facility Manager Support Desk	Attendant technical helpdesk and manual billing override auditing	Rs. 22,000	Rs. 2,64,000
Total	Operational Maintenance Budget	Rs. 1,02,000	Rs. 12,24,000

Combined Budget Summary

Category	Monthly Cost (Rs.)	Annual Cost (Rs.)
Risk Management & Mitigation Reserve	Rs. 55,000	Rs. 6,60,000
Operational & Hardware Maintenance	Rs. 1,02,000	Rs. 12,24,000
Total Financial Allocation	**Rs. 1,57,000**	**Rs. 18,84,000**